**Experiment No.:1**

Date of Performance:
Date of Submission:

**Aim:** Study of electromagnetic spectrum allocation and wireless bands used in communication.

**Required Equipment / Software** : Computer/Laptop,Internet connection ,Wi-Fi Analyzer / similar app

**Theory:**

The **electromagnetic spectrum** is the complete range of electromagnetic radiation arranged according to frequency or wavelength.

The approximate relationship between frequency and wavelength is : $f = c/\lambda$

where:

- c = speed of electromagnetic waves
- f = frequency
- $\lambda$ = wavelength

Therefore,

$$\lambda = c/f$$

**Wireless frequency bands** are specific ranges of radio frequencies allocated for different wireless communication services. Wireless systems transmit information by converting electrical signals into **electromagnetic radio waves**, which propagate through free space.

The choice of frequency band has a major effect on **communication range, bandwidth, antenna size, propagation, penetration, interference, and data rate**.

## Procedure

- Study the electromagnetic spectrum from low frequency to high frequency using a reliable reference.
- Record the frequency range, wavelength range and applications of the major spectrum regions.
- Study the wireless technologies listed in the observation table and record their frequency bands and applications.
- Prepare the result and conclusion based on experimental observations

**Observation Table A : Major Regions of the Electromagnetic Spectrum**

| Region | Frequency | Example Applications | Notes |
|---|---|---|---|
| Radio waves | Up to about 300 MHz | Broadcasting, communication | Long-wavelength radio services |
| VHF | 30–300 MHz | FM radio, TV, aviation | Part of radio spectrum |
| UHF | 300 MHz–3 GHz | TV, cellular, wireless systems | Widely used for mobile communication |
| Microwave | ~1–300 GHz | Radar, satellite, Wi-Fi, cellular | High-capacity wireless links |
| Infrared | ~300 GHz–400 THz | Remote controls, sensors, optical links | Short-range/line-of-sight applications |
| Visible light | ~400–750 THz | Imaging, optical communication | Visible to human eye |
| Ultraviolet | ~750 THz–30 PHz | Medical and industrial applications | Higher-energy radiation |
| X-rays | ~30 PHz–30 EHz | Medical imaging | Ionizing radiation |
| Gamma rays | Above ~30 EHz | Nuclear/medical applications | Highest-energy region |

**Observation Table B : Wireless Frequency Bands / Technologies to Study**

| Sr. No. | Technology | Typical Frequency | Application | Spectrum Type |
|---|---|---|---|---|
| 1 | NFC | 13.56 MHz | Contactless communication | ISM / short-range |
| 2 | FM Radio | 88–108 MHz (India) | FM broadcasting | Licensed |
| 3 | Cellular (2G/4G/5G) | Various licensed bands | Mobile communication | Licensed |
| 4 | GPS | ~1.575 GHz and other bands | Navigation | Satellite service |
| 5 | Bluetooth | 2.4 GHz ISM | Short-range/PAN | Unlicensed ISM |
| 6 | Zigbee | 2.4 GHz ISM | IoT / low-power PAN | Unlicensed ISM |
| 7 | Wi-Fi | 2.4 GHz | WLAN | Unlicensed/regulated |
| 8 | Wi-Fi | 5 GHz | WLAN | Unlicensed/regulated |
| 9 | Wi-Fi 6E/7 | 6 GHz where permitted | High-capacity WLAN | Regulatory dependent |
| 10 | Satellite / Radar | Various microwave bands | Satellite/radar links | Licensed/allocated |

**Result :**

The electromagnetic spectrum and frequency bands used for wireless communication were studied. The frequency, wavelength, applications and spectrum characteristics of different wireless technologies were analyzed.

## Conclusion:

The experiment demonstrates that wireless communication technologies operate in specific frequency bands selected according to propagation characteristics, capacity, interference considerations and regulatory requirements.

**Viva Voce Questions** :

1. What is the electromagnetic spectrum?
2. What is the relationship between frequency and wavelength?
3. What is an ISM band?
4. Which frequency band is commonly used by Bluetooth?
5. Which frequency bands are used by Wi-Fi?
6. What is spectrum allocation?
7. Why is spectrum allocation required?
8. What is licensed spectrum?
9. What is unlicensed spectrum?
10. What is the difference between 2.4 GHz and 5 GHz Wi-Fi?

**MARKS & SIGNATURE :**

| R1 (3 Marks) | R2 (5 Marks) | R3 (4 Marks) | R4 (3 Marks) | Total (15 Marks) | Signature |
|---|---|---|---|---|---|
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